

1. Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.



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Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config)#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
VLINE-S-CHANGED: Interface GigabitEthernet0/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#end
Router#
VSYN-5-COMFIG_I: Configured from console by console

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 EXSTART/DR 00:00:34 10.0.0.2 GigabitEthernet0/0
Router#
VSYN-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 FULL/DR 00:00:30 10.0.0.2 GigabitEthernet0/0
Router#
VSYN-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
```

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
VLINE-S-CHANGED: Interface GigabitEthernet0/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
VLINE-S-CHANGED: Interface GigabitEthernet0/1, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
VSYN-5-COMFIG_I: Configured from console by console

Router#
Router#
Router#
VSYN-5-ADJCHG: Process 1, Nbr 10.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
10.0.0.1 1 FULL/DR 00:00:37 10.0.0.1 GigabitEthernet0/0
20.0.0.2 1 EXSTART/DR 00:00:30 20.0.0.2 GigabitEthernet0/1
Router#
VSYN-5-ADJCHG: Process 1, Nbr 20.0.0.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router#
```

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router#enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
VLINE-S-CHANGED: Interface GigabitEthernet0/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
VSYN-5-COMFIG_I: Configured from console by console

Router#
Router#
Router#
VSYN-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 FULL/DR 00:00:36 20.0.0.1 GigabitEthernet0/0
Router#
VSYN-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
```

2. Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 EXSTART/DR 00:00:34 10.0.0.2 GigabitEthernet0/0
Router#
00:06:34: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 FULL/DR 00:00:30 10.0.0.2 GigabitEthernet0/0
Router#
Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 FULL/DR 00:00:34 10.0.0.2 GigabitEthernet0/0
Router#
Router#

Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#
00:06:22: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
10.0.0.1 1 FULL/DR 00:00:37 10.0.0.1 GigabitEthernet0/0
20.0.0.2 1 EXSTART/DR 00:00:30 20.0.0.2 GigabitEthernet0/1
Router#
00:06:38: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router#
Router#

Router2
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 20.0.0.0 0.0.0.255 area 1
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#
00:06:24: %OSPF-5-ADJCHG: Process 1, Nbr 20.0.0.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
20.0.0.1 1 FULL/DR 00:00:36 20.0.0.1 GigabitEthernet0/0
Router#
Router#
```

